Just In Time Adaptive Intervention

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Background: Why it matters

What is chronic stress?

- A prolonged state of feeling pressured or overwhelmed that lasts for weeks, months, or even years.
- Keeps the body's stress response activated over time, increasing the risk of poor mental and physical health.

Current Solutions

- Psychological treatments (e.g., therapy)
- Lifestyle changes (e.g., exercise, sleep, relaxation)
- Medical treatments (e.g., medication)

The Gap

- Most stress apps rely on self-reported feelings or fixed schedules, missing the moments when support is needed most (Teepe et al., 2021)
- Traditional care like therapy can be costly or difficult to access, while affordable, accessible, timely support remains an unmet need

The Opportunity: Timely Micro-Interventions

Deliver brief, personalized support at the moments when stress begins to rise, helping people manage stress before it builds up.

Real-Time Stress Detection & Intervention

Can wearable signals detect stress and trigger timely micro-interventions?

Stress Detection

Use HRV, heart rate, EDA, and movement to detect potential stress in near real time.

(Pinge et al., 2024; Bögemann et al., 2025)

Timely Intervention

Offer brief support—such as paced breathing or mindfulness—when stress is detected.

*(*Effectiveness of 2 Just-in-Time Adaptive Interventions*, 2025)*

Evidence

Early studies show that real-time stress detection and sensor-triggered interventions are feasible.

(Bögemann et al., 2025)

Key Challenge

Separate stress from exercise or other physiological changes; test whether adaptive timing improves outcomes.

*(Pinge et al., 2024; *Effectiveness of 2 Just-in-Time Adaptive Interventions*, 2025)*

Current Model Example: WESAD

A classic baseline for wearable-based stress detection

INPUT

ECG / heart rate
EDA · temperature
respiration · movement

→

MODEL

Extract features from sensor windows → SVM or Random Forest

→

OUTPUT

Stress vs. non-stress classification per window

Reference: Schmidt et al. (2018), "Introducing WESAD: A Multimodal Dataset for Wearable Stress and Affect Detection."

Key Details

- 15 participants; controlled lab study
- Useful baseline, not a validated real-world detector
- Real-time use needs live sensor streaming and testing on new users

System/Analysis: Data & Features

Data & Sensors: WESAD dataset

- **Data:** 15 subjects; labeled baseline, stress, and amusement conditions
- **Chest (RespiBAN):** ECG, EDA, EMG, respiration, skin temp, 3-axis accelerometer
- **Wrist (Empatica E4):** BVP (PPG), EDA, skin temp, 3-axis accelerometer
- **Storage:** ~2 GB (confirm after download), stored locally with no server
- **Pipeline:** load each subject's file in Python, align signals with labels, save cleaned windows and features as Parquet; back up to Google Drive, code on GitHub

Source: Schmidt et al., ICMI 2018 | WESAD dataset page

Data issues & handling

- **Motion artifacts** in EDA/PPG: filtering and accelerometer-based rejection
- **Missing data:** interpolation or window exclusion
- **Label noise:** lab labels differ from real-world self-report
- **Small, narrow sample (bias):** check subgroup results (slide 7); state limits for groups WESAD lacks

Feature engineering



System/Analysis: Inference & JITAI Logic

Inference: baseline classifiers

Classify baseline / stress / amusement:

- **Models:** logistic regression (ref), random forest (Schmidt et al.)
- **Why classical ML:** 15 subjects is small; deep learning (PyTorch) only as a stretch
- **Train/test:** leave-one-subject-out; tune hyperparameters on inner folds; fit normalization and scaling on training subjects only; held-out subject untouched
- **Stretch goal:** personalization

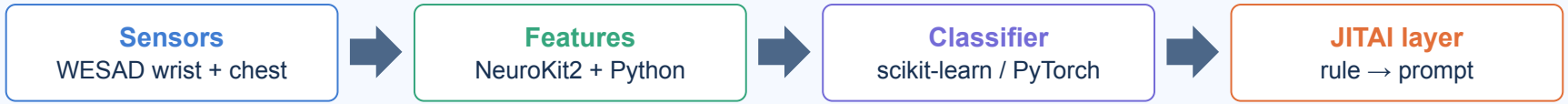
JITAI decision logic

A rule-based triggers an intervention only when all hold:

- **Stress is sustained:** predicted stress probability stays high across recent windows
- **Person is receptive:** e.g., not exercising (low accelerometer activity)
- **Not over-prompted:** cooldown between prompts and a daily cap

Then deliver a brief micro-intervention, such as paced breathing.

End-to-end flow & tools



Validation Plan

1 Model metrics

Primary task: stress vs. non-stress, leave-one-subject-out CV

F1
Balances precision and recall

Balanced accuracy
Handles uneven classes

AUROC
Separation at any threshold

2 JITAI metrics

Does the trigger fire at the right moments?

Precision
False alarms cause fatigue

Latency
Stress onset to trigger

User study
~8-10 people rate timing and helpfulness (if feasible)

Success criteria

AUROC > 0.80
Quantitative

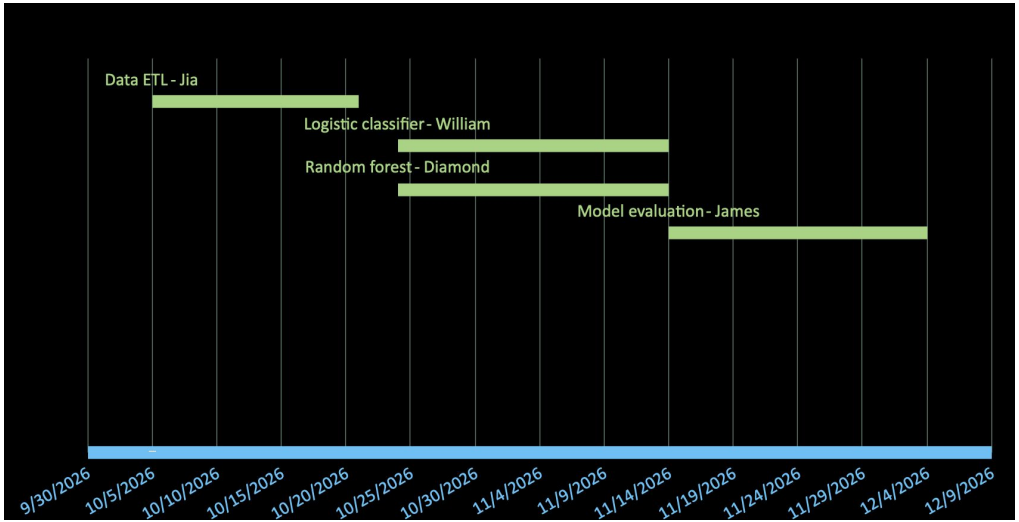
F1 > baseline
Quantitative

Majority
Qualitative: prompts well timed (if user study is run)

Explainability & Interpretation

Which signals drive predictions?	**SHAP** and **permutation importance** rank the signals, e.g., EDA vs. HRV	OUTPUT **Ranked importance plot**
Does it work for everyone?	Performance by **gender, age**, and **sensor placement**. Skin tone (PPG accuracy) and economic background are not in WESAD, so we state them as limits	OUTPUT **Metrics by group**
Can we trust the data?	**Motion-artifact rates** and **per-subject performance variance** show where results are weak	OUTPUT **Artifact and variance table**

Milestones



Potential roadblocks

- **Data format:** WESAD ships as Python pickle files; budget time for loading and syncing signals.
- **Noisy signals:** motion artifacts may need extra cleaning; keep a buffer after ETL.
- **Small sample:** leave-one-subject-out results may vary a lot; report per-subject scores.
- **User study:** recruiting 8-10 testers may slip; fallback is replay-only evaluation.
- **Team schedule:** other coursework in November; weekly check-ins to catch delays early.

References

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